

SWARAJ ZENDE

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EXPERIENCE

Addverb Technologies — Robotics Engineer

September 2024 – Present

- Owned the end-to-end software stack of **Zippy (AGV)** across legacy and upcoming global deployments, acting as the primary point of contact for system stability, feature evolution, and hardware integration.
- Optimized the AGV motion controller, improving goal-reaching accuracy from **7 cm to 1 cm** through control-loop tuning and parameter optimization.
- Redesigned the communication architecture to reduce system latency and CPU load, lowering utilization from **95% to 20%** by replacing heavy ROS middleware with optimized custom drivers.
- Established a hybrid runtime architecture by isolating ROS strictly for debugging and observability, significantly improving real-time determinism and runtime reliability.
- Designed modular **Low-Level Drivers (LLDs)** and hardware abstraction layers (IO handlers, attachment handlers), enabling backward compatibility with older hardware revisions while simplifying integration of new hardware variants.

IIT Bombay — Summer Intern

May 2024 – July 2024

- Built a full-cycle warehouse automation workflow coordinating a robotic arm and an AMR, integrating **Nav2** for autonomous navigation and **MoveIt** for manipulation planning within a unified ROS2 system.
- Replaced monolithic state machines with **Behavior Trees**, improving modularity, fault isolation, and ease of extending complex interaction sequences between navigation and manipulation subsystems.

Sakar Robotics — Robotics Software Intern

December 2023 – May 2024

- Designed a modular software architecture integrating a differential-drive Pick-Assist robot with the **Nav2** stack.
- Developed ground-up ROS drivers bridging high-level navigation/control nodes with low-level embedded motor controllers.
- Validated navigation and control on simulation and hardware, tuning parameters and resolving cross-layer integration issues.

MAJOR PROJECTS

Multiplayer Chess Game

C++ — SFML

- Developed a full-stack multiplayer chess application using a custom **state-machine architecture** to manage menu navigation, gameplay flow, and analysis modes.
- Implemented a single-threaded, non-blocking event loop to maintain responsive gameplay while concurrently processing user input, rendering, and network communication.
- Built an analysis board with a built in Backtracking chess engine with Alpha Beta Pruning for further Optimization.

LightYears (Space Shooter Game)

C++ — SFML

- Designed a state-driven game architecture to manage transitions between menus, gameplay, and combat states in a clean and maintainable manner.
- Implemented a multi-tier enemy AI system with progressively increasing difficulty, encapsulating movement, firing patterns, and behavior logic per enemy type.
- Built a modular weapon framework that supports multiple weapon types and power levels, enabling extensibility without modifying core gameplay logic.
- Integrated a probabilistic reward system that dynamically grants lives and power-ups based on enemy difficulty and game progression.

POSITIONS OF RESPONSIBILITY

Software Domain Head — VIIT Robotics Club

July 2022 – July 2024

- Led the software domain for a 30-member robotics team, coordinating a 6-member subgroup across navigation, control, and system integration.
- Drove the team to qualify for **Stage 3** of ABU Robocon 2023, securing an **All India Rank (AIR) 21**.

EDUCATION

B.Tech in Electronics & Telecommunication — VIIT, Pune CGPA: 8.53 / 10

July 2024

SKILLS

Languages: C, C++, Python

Robotics & Control: Closed-Loop Control, Controller Design & Tuning, Trajectory Tracking, Differential Drive Kinematics, Sensor Fusion, TF2

Systems & Middleware: ROS1/ROS2, Nav2, BehaviorTree.CPP, CMake, Git, Docker

Industrial Communication: EtherCAT, CAN, Serial Interfaces